



OUTLIERS

Capture

Layer 4 of 8 · Where the typing stops,
and the system starts filling itself.

FOR MAC

EST. MMXXVI · OUTLIERS ACCELERATOR

It fills itself

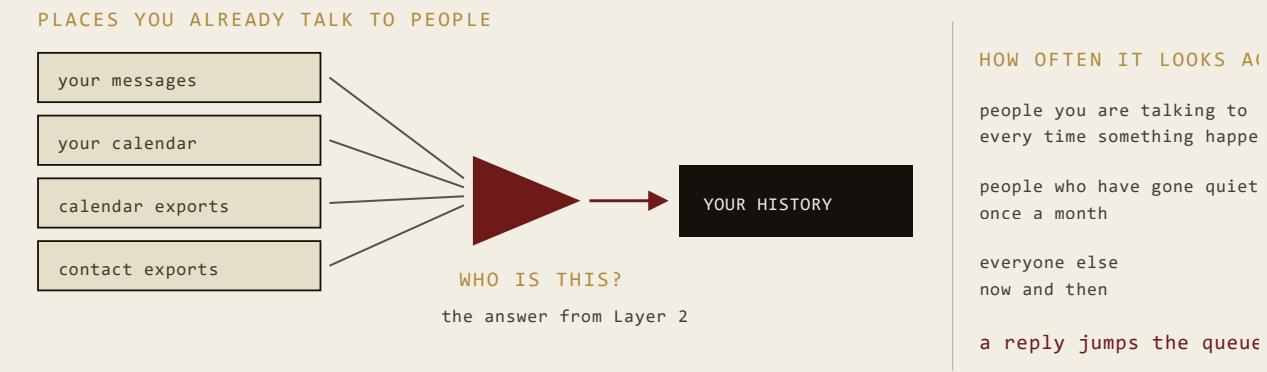
Everything so far has been groundwork. This is the layer where your CRM starts keeping a history you did not type.

WHAT THE LAST LAYER LEFT YOU WITH

You have a place to record what happened and rules for working out everything else from it. What you do not have is anything putting the history in, which leaves you typing, which is what kills CRMs.

WHAT THIS LAYER BUILDS

Small programs that read an export from a place your conversations already happen and turn every row of it into history. You run one command against the file, and nothing is typed. Running the same export again adds nothing twice, so you can drop a newer one on top whenever you like.



EVERY SOURCE GOES THROUGH THE SAME QUESTION BEFORE ANYTHING IS WRITTEN DOWN.

How it works

Collecting the records is straightforward. Working out which person each one belongs to is not, and that is most of this layer.

WORKING OUT WHO EACH RECORD BELONGS TO

Getting a list of messages out of somewhere is straightforward. The difficulty is that the same human turns up as a profile in one place, an email address in another, and a first name on a meeting invitation in a third, while 2 different people share a name.

So everything a collector finds goes through the same question first, using the answer you built in Layer 2: **who is this?** Nothing is written down until that is settled. A record filed against the wrong person corrupts every calculation above it, and silently, which makes it far worse than a record not filed at all.

When your system cannot tell who something belongs to, it records it anyway and marks it unattributed. Throwing it away would leave the history looking complete when it is not.

NOT EVERYTHING NEEDS LOOKING AT AS OFTEN

You cannot re-check everything constantly, and checking nothing lets the whole CRM go stale. So contacts sit in 3 groups. People you are actively talking to are refreshed whenever something happens, which costs nothing extra. Warm contacts get looked at occasionally. Everyone else, rarely.

Anything that actually happens overrides that schedule: a reply is picked up straight away rather than waiting its turn.

NOTHING HERE USES AI

These are small, boring programs that do the same job every time and can be tested by giving them a known input and checking the output. That is deliberate. You want the programs that decide what is true in your system to be predictable. Judgement arrives in the next layer, on top of this, where you can see it and check it.

Before you start on a Mac

Everything in this guide is typed into **Terminal**, the Mac's window for typing commands instead of clicking. Press Command and Space together, type Terminal, and press Return.

This layer needs Python and git, which you set up for Layer 1 (its page "Before you start on a Mac"), and the layers before this one. It installs no add-ons, so it works with either Python a Mac may have.

HOW TO TYPE THE LINES

Where this guide shows several lines in a black box, type them one at a time: type the first, press Return, wait for it to finish and give you a fresh line, then type the next. The installer asks you questions. Each offers an answer in [square brackets]; press Return to take it, or type your own and press Return.

WHERE YOUR CRM IS

Layer 1 made your CRM at ~/CRM: a folder called CRM in your home folder (the folder with your name and the house picture in Finder's sidebar). If you chose another folder in Layer 1, put your own folder wherever a line in this guide says ~/CRM. The installer finds your CRM by itself either way.

Apple Silicon or Intel (the 2 kinds of chip a Mac can have; the Apple menu, then About This Mac, shows yours): the steps are the same on both, and both were tested.

Tried only on test Macs. Every step in this guide and in the instructions file of the layer's download was tried on test Macs (Macs GitHub rents out by the minute to run a script, not a person's own Mac, and not yours), never on a real Mac, except 3: the box asking to install the developer tools, which cannot appear on a test Mac; `python3 _engine/collect.py run connections <file.csv> --dry-run`, because it is a pattern: you type your own export file where is; `python3 _engine/collect.py run connections <file.csv>`, because it is a pattern: you type your own export file where is. The last pages list every line and what happened to it.

Build it

3 lines, typed into Terminal. The last one asks you a few questions so what you get is yours, then does the work.

```
git clone https://github.com/OUTLIERS-ai/outliers-crm-04-capture-mac
cd outliers-crm-04-capture-mac
python3 install.py
```

This layer needs Layers 1 to 3. It checks first and tells you if anything is missing.

WHAT IT ASKS YOU, AND WHY

QUESTION	WHAT IT CHANGES
Where do your conversations actually happen?	Messages, email, calls, meetings, a community. Pick the ones that matter; you can add more later.
Which of those can you export or connect?	Most platforms let you download your own data. The installer walks you through the ones you pick, and skips anything you cannot get at yet.

WHAT YOU END UP WITH

- A collector for each source you connected, each writing into your one history.
- Every incoming item passed through the same identity check before it is filed.
- A refresh schedule so your records are re-checked in proportion to how much they matter.

Now use it

Connect one source, and read a history you did not type.

Connect one source. Just one, the easiest one you have.

Then open the record of someone you have not thought about in a while and read their history. Depending which source you chose, you may find exchanges you had forgotten and meetings you had lost track of. The system records that something happened and when, not what was said in them.

Nobody typed any of it. That is what the first 3 layers were for.

IF YOU ALREADY HAD SOMETHING LIKE THIS

Your test: pick a person and count how many separate places their history currently lives. If the answer is more than one, you have several partial records of the same relationship and no single view of it. That is what this layer removes.

WHAT THIS LAYER LEAVES UNSOLVED

Your system can now tell you what happened. It cannot tell you what any of it means. Whether this person is worth an hour of your week, whether a reply is warm or merely polite, and what to say back are all judgements, and a small program cannot make them.

NEXT: LAYER 5 • JUDGEMENT

Assistants that read, score and draft, each with one job, each showing where every fact it writes down came from.

WORDS USED IN THIS GUIDE

Collector	A small program that watches one source and records what happens there.
Attribution	Deciding which person something belongs to.
Unattributed	Something recorded that the system could not confidently assign to anyone.
Refresh	Looking again at details that may have changed, such as a job title.

What was run on a Mac

A **test Mac** is one of the Macs GitHub rents out by the minute to run a script.

Each line below is one this guide or the instructions file of the layer's download (README.md) gives you to type.

The run was on 2026-09-26. A script (a list of commands each test Mac carries out by itself) downloaded a fresh copy of the download from GitHub on each of 7 test Macs and ran each line below exactly as printed, in this order. After a `cd` line (`cd` moves Terminal into a folder), the lines that follow ran in that folder, as they do for you. The script ran them; no person typed them.

7 test runs: 3 on Macs set up from scratch (macOS 15.7.9 on Apple Silicon, macOS 15.7.9 on Intel, macOS 26.6.2 on Apple Silicon), and 4 on 2 Macs that also have Homebrew, an add-on installer many Mac owners use, each run once with Homebrew installed before Python and once after (macOS 15.7.9 on Apple Silicon and macOS 15.7.9 on Intel). Before the first line, the script put these on every Mac: Python from python.org; Layers 1 to 3 of the CRM, installed from their Mac downloads. GitHub's Macs already have Apple's developer tools, which include git.

EACH LINE, THEN WHAT HAPPENED TO IT

```
git clone https://github.com/OUTLIERS-ai/outliers-crm-04-capture-mac
```

worked on all 7 test Macs

```
cd outliers-crm-04-capture-mac
```

worked on all 7 test Macs

```
python3 install.py
```

worked on all 7 test Macs

```
cd ~/CRM
```

worked on all 7 test Macs

```
python3 _engine/collect.py list
```

worked on all 7 test Macs

```
python3 _engine/collect.py run connections <file.csv> --dry-run
```

not run in this test: it is a pattern: you type your own export file where <file.csv> is

```
python3 _engine/collect.py run connections <file.csv>
```

not run in this test: it is a pattern: you type your own export file where <file.csv> is

```
python3 _engine/collect.py all
```

worked on all 7 test Macs

EACH LINE, CONTINUED

```
python3 _engine/refresh.py due
worked on all 7 test Macs

python3 _engine/refresh.py tiers
worked on all 7 test Macs

python3 _engine/refresh.py checked "a name"
worked on all 7 test Macs

python3 _engine/ledger.py person "a name"
worked on all 7 test Macs

python3 _engine/derive.py quiet 60
worked on all 7 test Macs

cd ~/outliers-crm-04-capture-mac
worked on all 7 test Macs

python3 tests/test_capture.py
worked on all 7 test Macs

python3 tests/test_refresh_tiers.py
worked on all 7 test Macs

python3 tests/test_atomic_writes.py
worked on all 7 test Macs
```

The script's record of the run is public, for anyone who wants to check it; you do not need to open it:
<https://github.com/OUTLIERS-ai/outliers-mac-test/actions/runs/36210558078>

NOT RUN ON THE TEST MACS

The box asking to install the developer tools cannot appear on a test Mac, because GitHub's Macs already have them.

Opening your CRM in Obsidian. The test Macs have no Obsidian; the script read the folders and files the installer made instead.